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PQR: Generating Diverse & Realistic User Queries that Elicit QA Agent Failures



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PROBLEM

Evaluating LLM agents demands **costly human effort** to design realistic failure scenarios. Prior adversarial methods generate **arbitrary, often gibberish** failure queries with no real user intent.

Realistic user queries — not just adversarial noise — break agents too, yet prior work overlooks them.

MOTIVATION

- **Adversarial jailbreaks** (PAIR, TAP) rely on predefined strategies and yield unrealistic queries of limited diagnostic value.
- **Prompt-refinement** (GEPA, ACE) optimizes effectiveness but ignores **query diversity** — repeatedly eliciting the same failure reveals little.

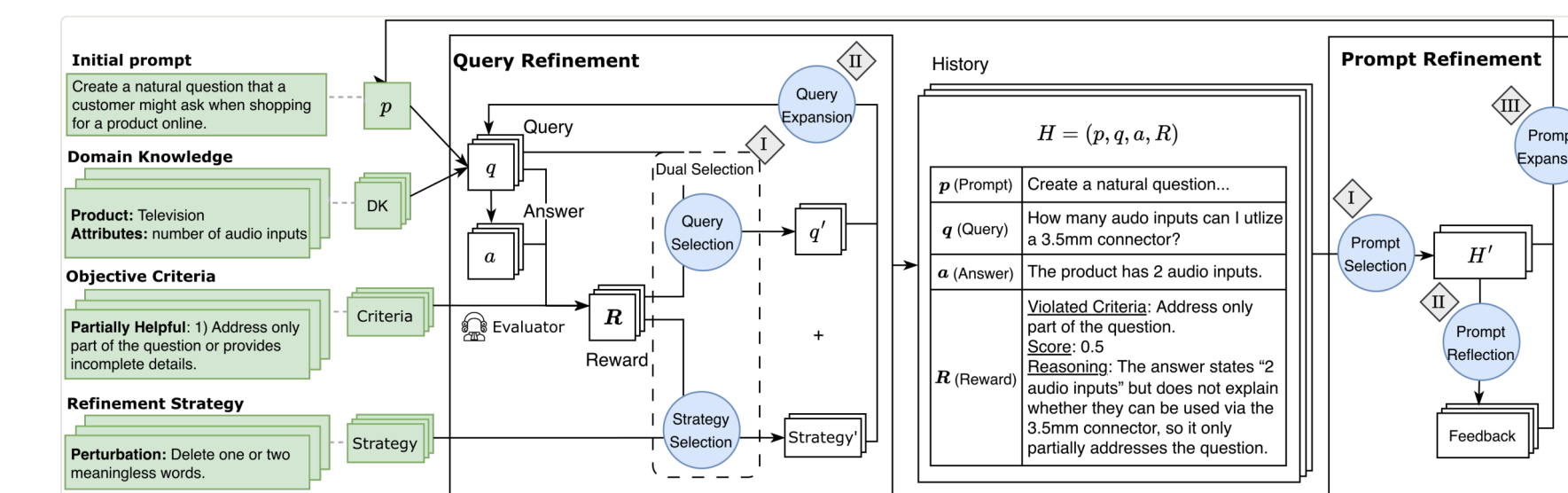
PAIR TAP GEPA ACE PQR (ours)

TAKEAWAY

Query refinement alone yields diverse but unrealistic variants; prompt refinement alone yields realistic but repetitive queries. Coupling them lets PQR surface agent failures that are **both diverse and realistic**, going beyond hand-crafted test cases and arbitrary adversarial noise.

A scalable, black-box, agent-agnostic path to evaluating deployed LLM agents.

METHOD



Two coupled modules refine queries and prompts in one iterative loop.

1 Query Refinement diverse rewrites

- **Dual selection** runs query & strategy selection in parallel via beam search over an LLM-as-a-Judge reward.
- Retains only strategies that steer queries toward a **lower reward**, meaning more agent failures.

char / word / sentence perturb persona

tone scenario

2 Prompt Refinement Loop feedback-driven

- 1 **Aggregate** interaction history (p, q, a, R) across all prompt candidates.
- 2 **Select** prompts that most induce failures; **reflect** on realism + objective gaps.
- 3 **Expand** into new candidate prompts that generate realistic violations.

3 Guided Reflection realism + objective

Realism feedback explains why generated queries appear unnatural and how to make future ones more human-like.

Objective feedback adds three complementary signals that diagnose agent behavior.

Violation — features that trigger failures · **Compliant** — features that satisfy the objective · **Criterion-specific** — under-explored criteria.

4 Parallel Expansion three directions

Each iteration generates **one prompt per direction** — all share the realism feedback but draw on a different objective signal to widen strategy coverage:

- 1 **Exploitation** — uses violation feedback to amplify strategies that already triggered agent failures.
- 2 **Exploration** — uses compliant feedback to introduce contrasting strategies that push toward new failure modes.
- 3 **Examination** — uses criterion-specific feedback to probe the least-frequently-violated criteria.

Diverse prompts yield diverse queries, breaking the single-prompt bottleneck of prior prompt-refinement methods.

KEY RESULTS

METHOD	UHR ↑	COST ↓	REALISM ↑
GEPA	45.60%	×1.3	3.81
ACE	39.58%	×1.1	2.66
TAP	33.38%	×1.6	2.31
PQR (ours)	56.32%	×1	3.66

+23%-78% more unhelpful responses uncovered vs. baselines — at the lowest token cost (×1).

SO WHAT → PQR posts the highest unhelpfulness rate at the lowest cost while ranking top-2 on every realism & diversity metric.

HEADLINE NUMBERS

56.32% **Unhelpfulness rate (best)**
most efficient failure discovery

×1 token cost (lowest)
2.50 human realism (best)
100 products, 4 categories

SCAN TO READ



Paper